

CHEMICAL BONDING

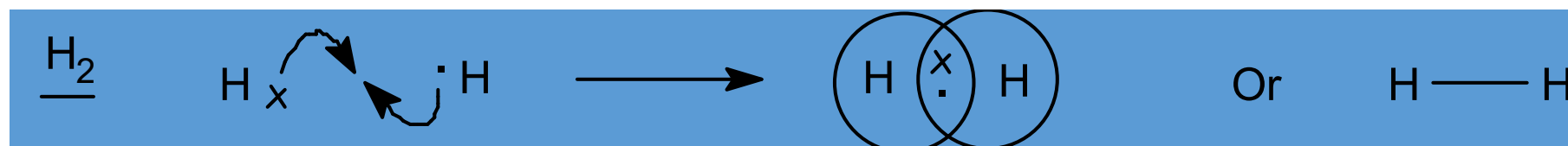
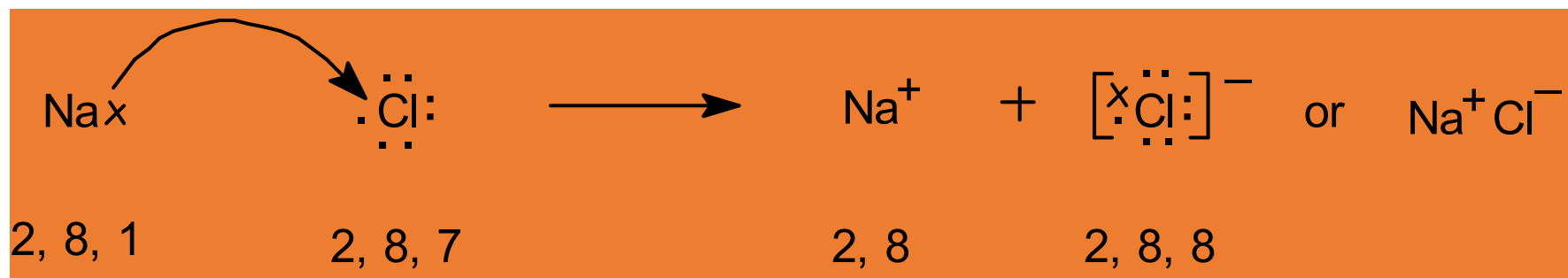
Text

**Introduction to Modern Inorganic
Chemistry**

By S. Z. Haider

CHEMICAL BOND

- **Definition:** A chemical bond is defined as a strong attractive force that exists between two or more atoms to hold them together as a stable molecule in a substance.



Some definitions to understand chemical bonding:

- **Valence electron**: The electrons in the outer most energy level in an atom that takes part in chemical bonding are called valence electrons.
- **Bonding electrons**: The valence electrons actually involved in bond formation are called bonding electrons.

Octet Rule

In chemical bond formation, atoms interact- (i) by losing, (ii) by gaining or (iii) by sharing electrons to acquire a stable noble gas configuration.

The tendency for atoms to have eight electrons in the outer shell is known as Octet rule.

Octet Rule (contd.)

He (2) 2

Ne (10) 2, 8

Ar (18) 2, 8, 8

Kr (36) 2, 8, 18, 8

Xe (54) 2, 8, 18, 18, 8

Rn (86) 2, 8, 18, 32, 18, 8

Main types of bond

- **Ionic or electrovalent bond**
- **Covalent bond**
- **Coordinate covalent bond**

Other important bonds:

- ✓ **Metallic bond**
- ✓ **Van der Waals bond**

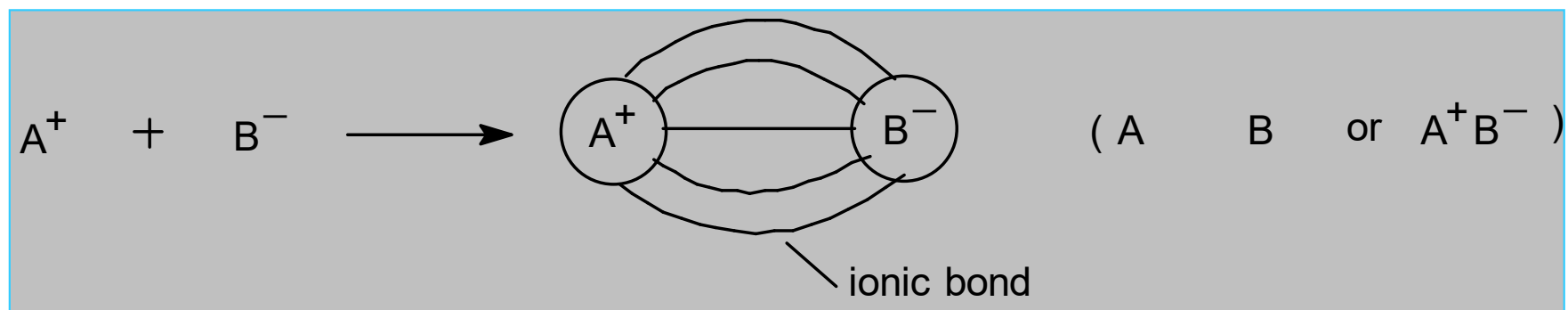
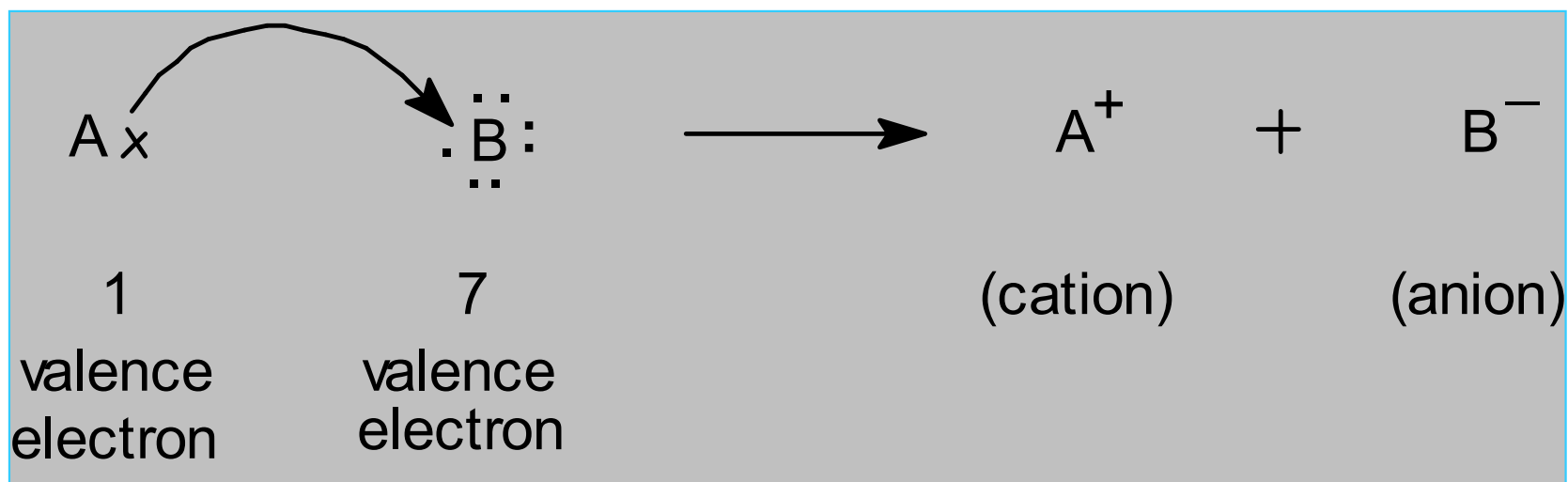
IONIC BOND

Definition: An ionic bond is *a chemical bond formed by the electrostatic attraction between positive and negative ions.*

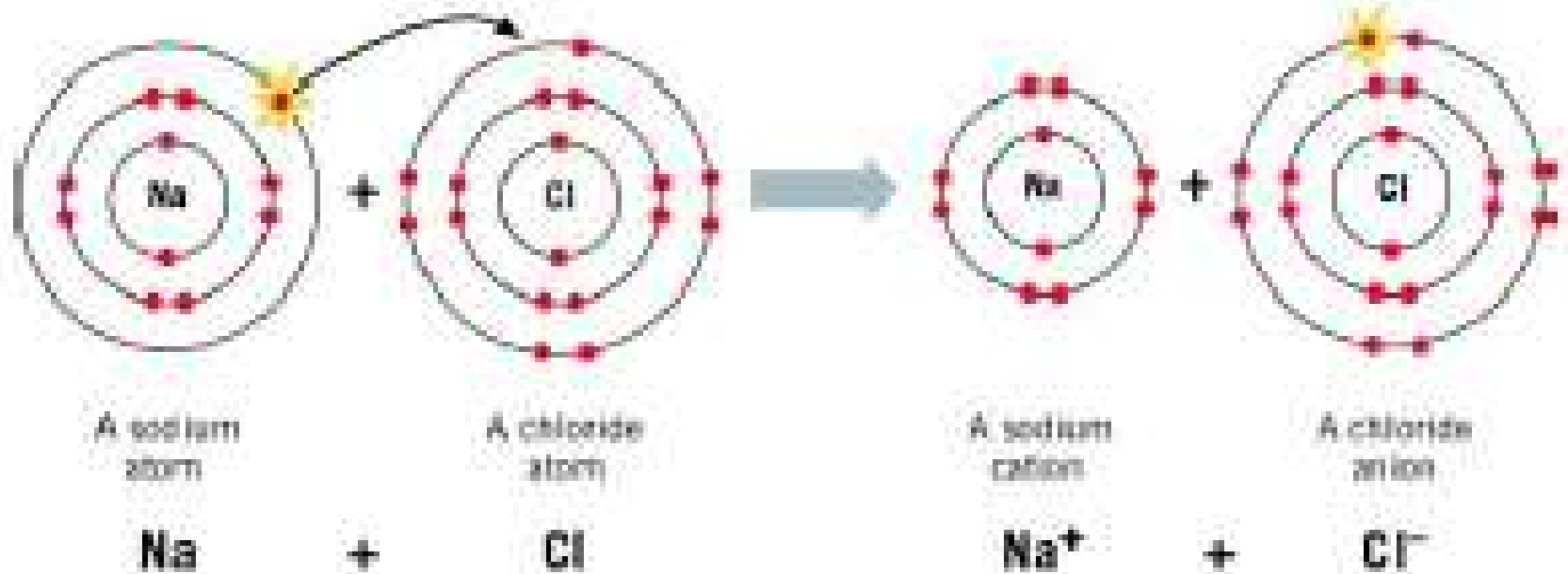
The bond forms between two atoms when one or more electrons are transferred from one atom's valence shell to the other's valence shell.

The atom that loses electrons becomes a *cation* (positive ion), and the atom that gains electrons becomes an *anion* (negative ion).

Ionic bond (contd.)

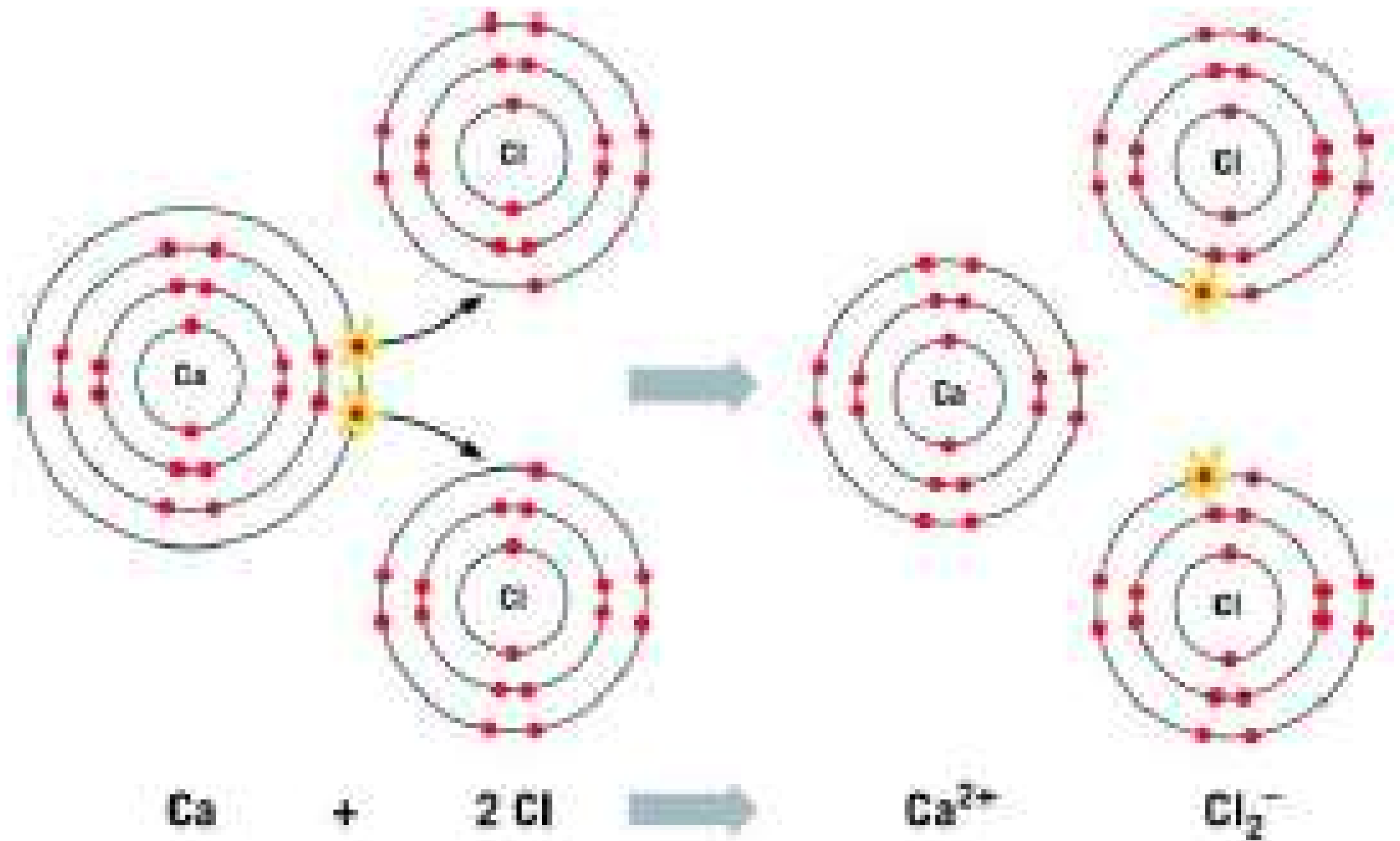


Ionic bond (contd.)



1. One atom gives up an electron to the other.
2. Each atom now has a full outer shell of electrons.
3. They are held together by opposite electrical charges.

Ionic bond with two electrons transfer



Conditions for formation of ionic bond:

- **Number of valence electrons:** The atom should have valence electrons.
- **Net lowering of energy:** To form a stable ionic compound, there must be a net lowering of the energy. For example-formation of NaCl molecule

Conditions for formation of ionic bond:



Net energy released, $187 + 85 - 119 = 153 \text{ kcal}$.

This causes lowering of energy of the process, hence ionic bond formed.

Conditions for formation of ionic bond:

- **Electro negativity difference between the atoms.**

Na $\rightarrow$ 0.9

Cl $\rightarrow$ 3.0

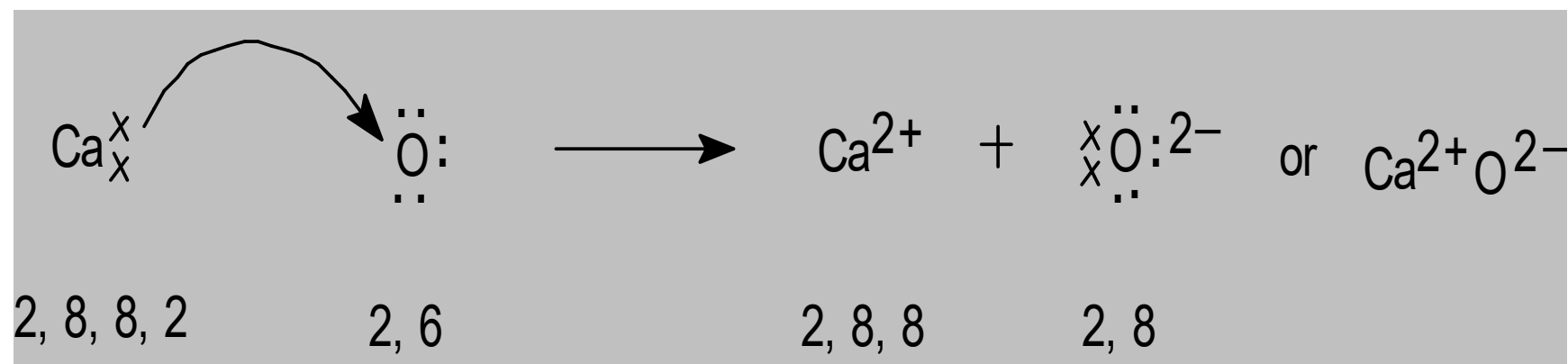
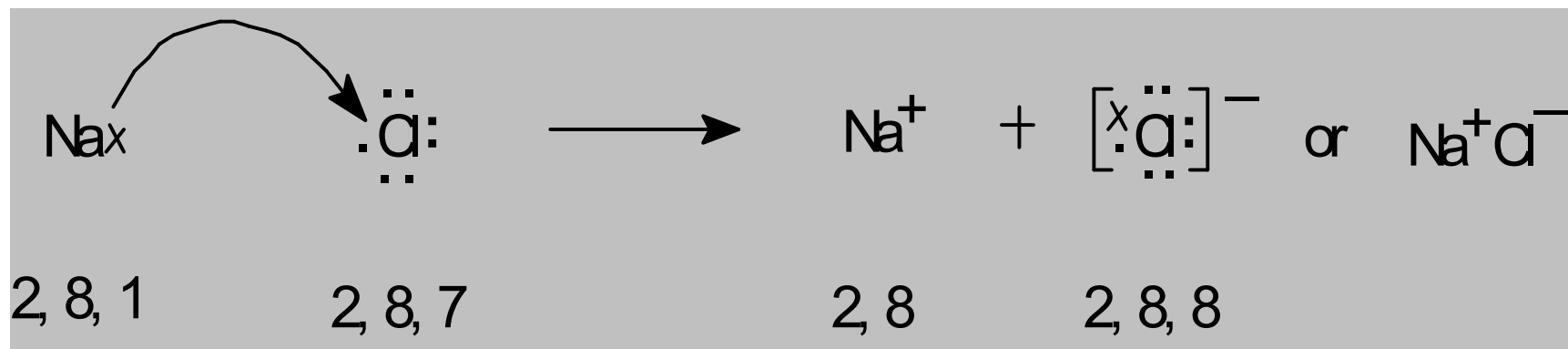
Difference, $3.0 - 0.9 = 2.1$

Hence ionic bond will form.

Factors that govern the formation of ionic bonds:

- Ionization energy**
- Electron affinity**
- Lattice energy**

Some examples of ionic bonds:

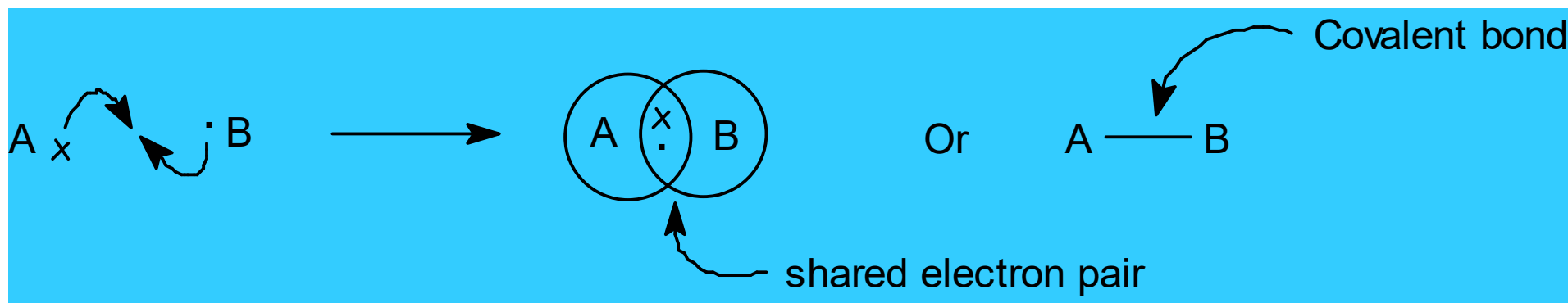


Characteristics of ionic compounds:

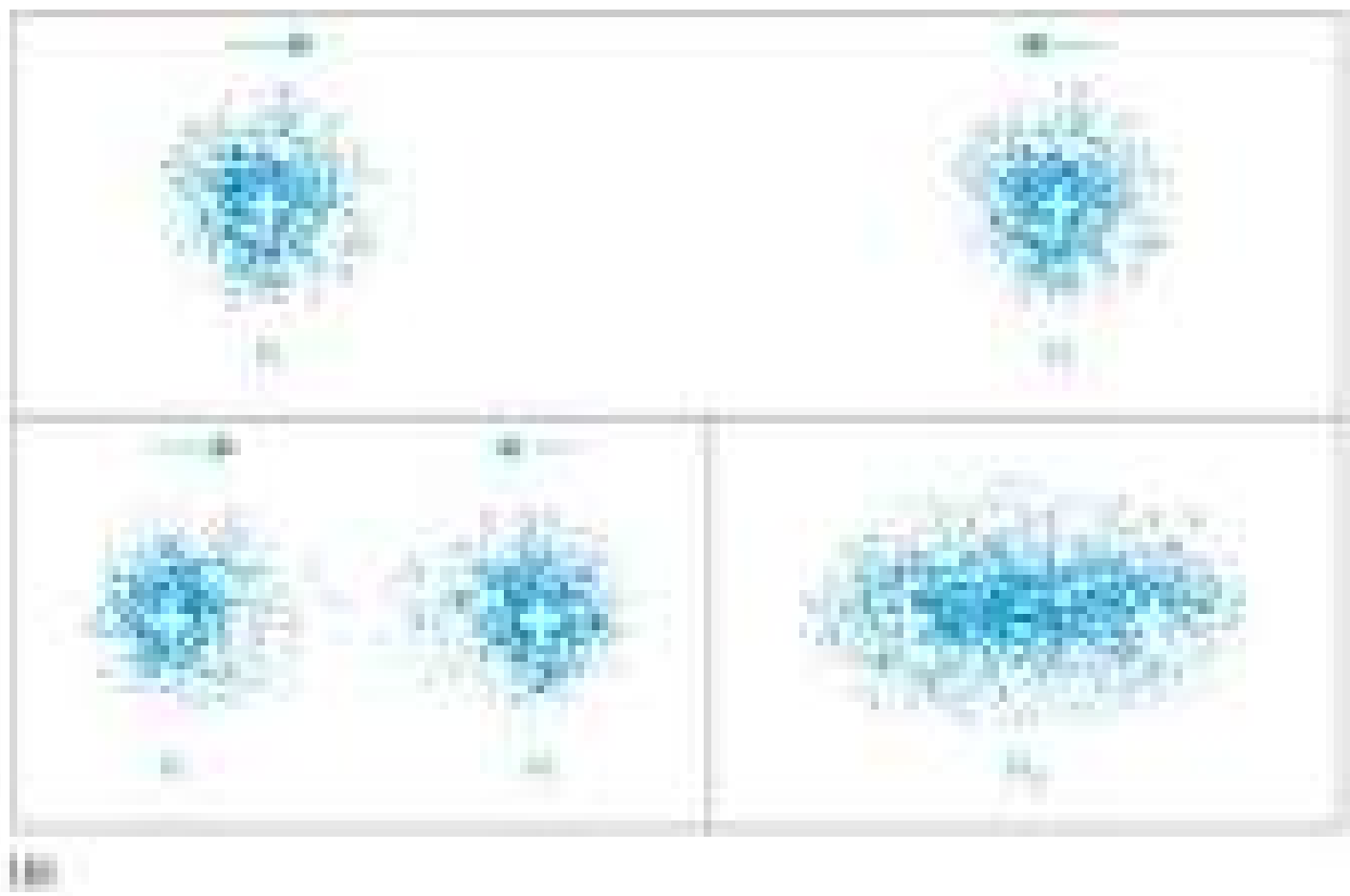
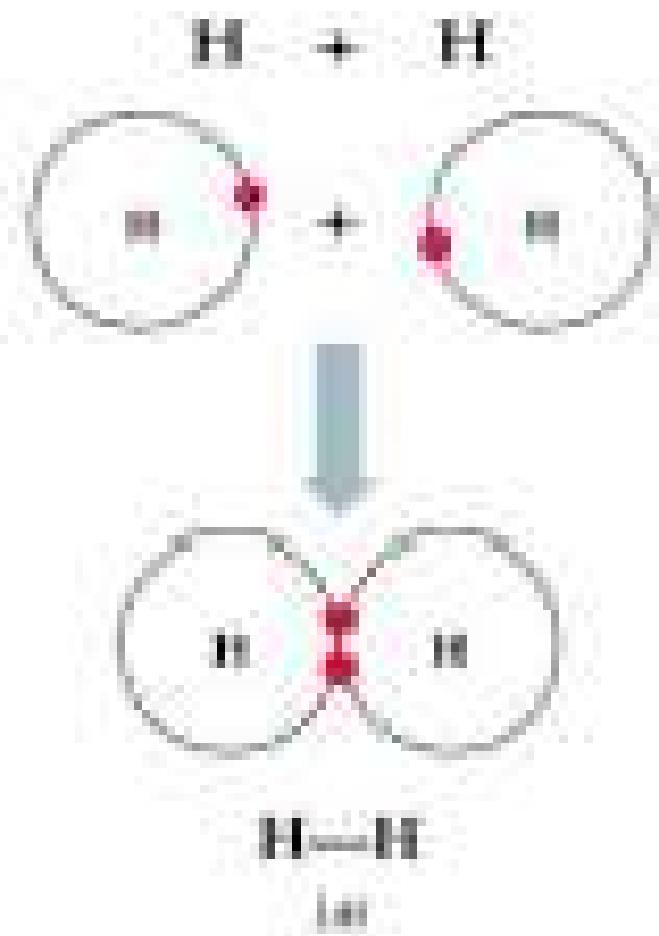
- solid at room temperature**
- high melting points**
- hard and brittle**
- soluble in water**
- conductors of electricity**
- don't exhibit isomerism**
- ionic reactions are fast**

COVALENT BOND

- **Definition:** The attractive force between atoms created by sharing of an electron-pair.



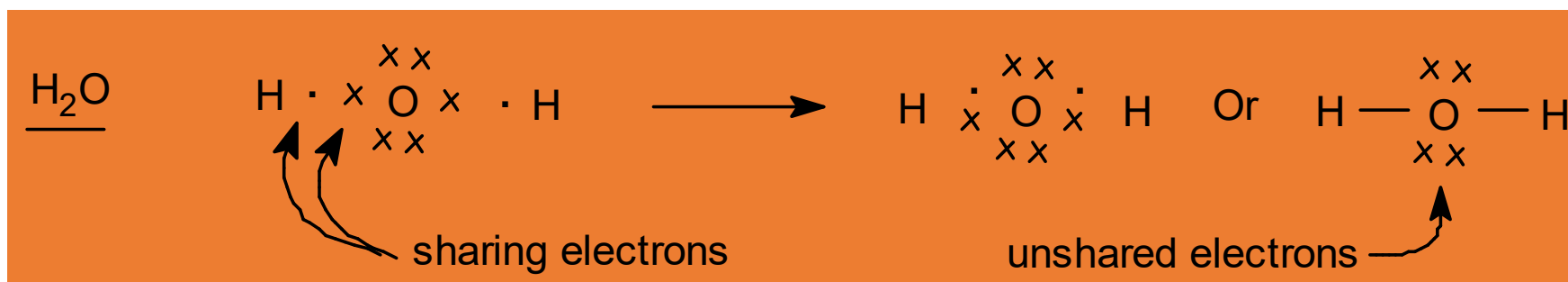
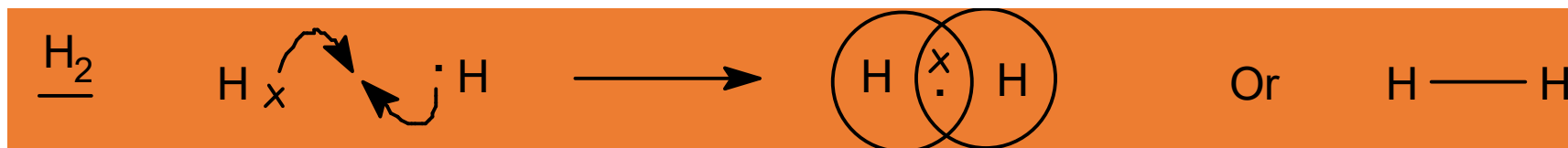
COVALENT BOND



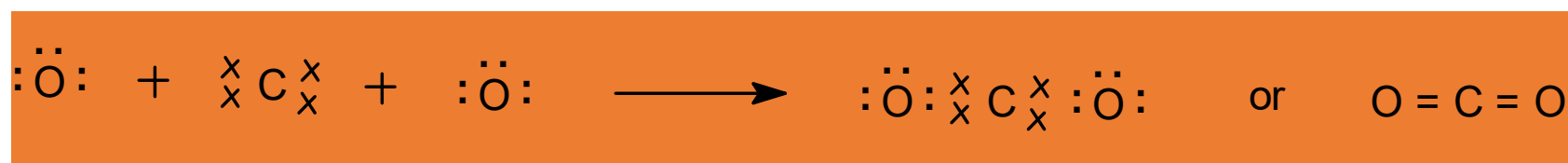
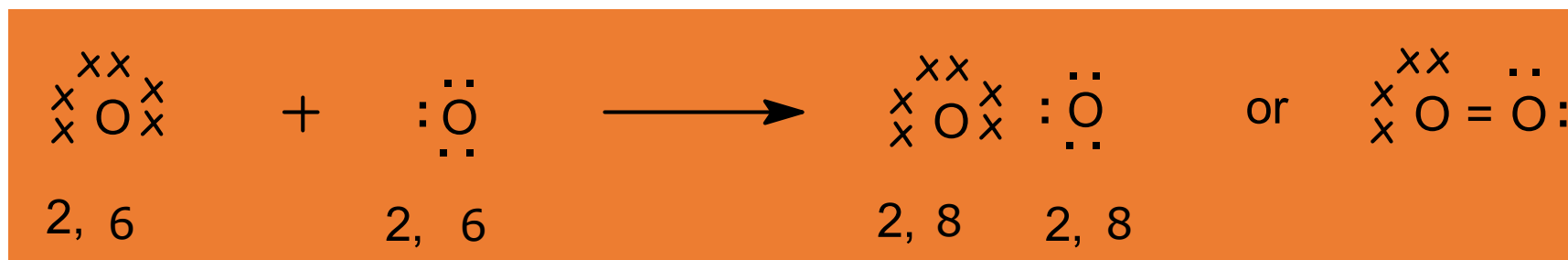
Conditions for covalent bond:

- **Number of valence electrons**
- **Equal electronegativity**
- **Equal sharing of electrons**

Examples of covalent compounds:



Examples of multiple covalent bonds:



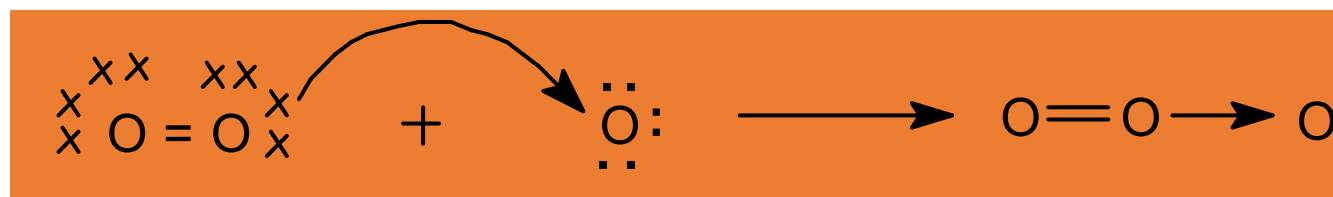
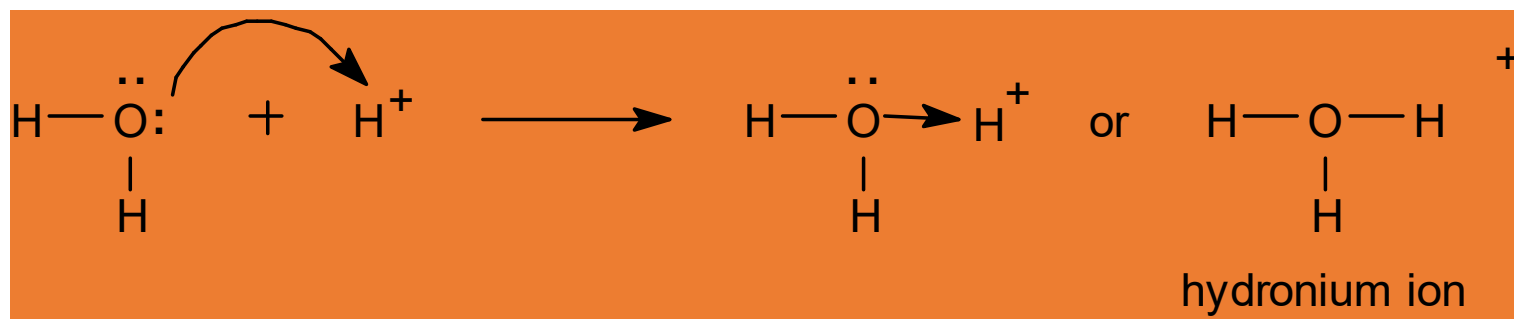
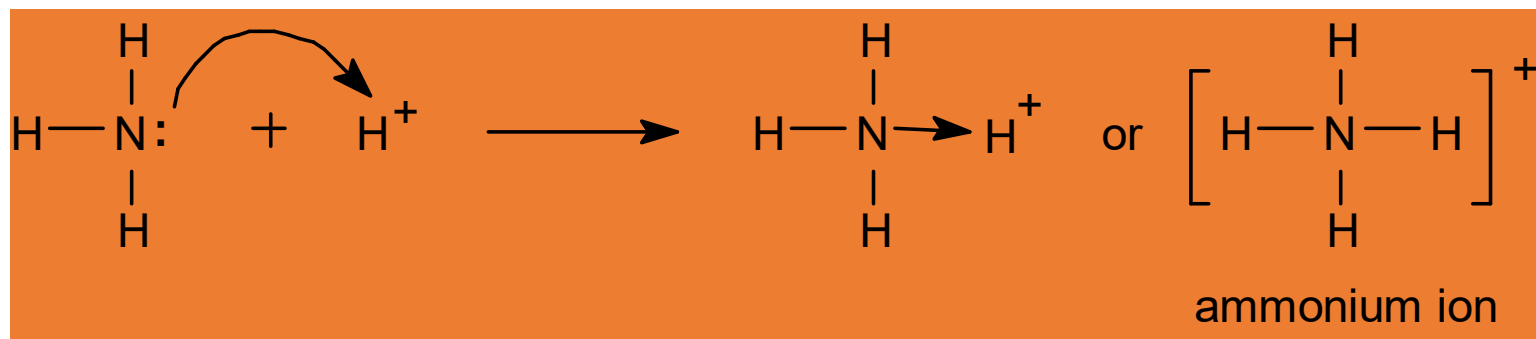
Characteristics of covalent compounds

- **Gases, liquids or solids at room temperature**
- **Low melting points and boiling points**
- **Neither hard nor brittle**
- **Soluble in organic solvents**
- **Non-conductors of electricity**
- **Exhibit isomerism**
- **Covalent reactions are slow**

COORDINATE COVALENT BOND

- **Definition: A covalent bond in which both electrons of the shared pair come from one of the two atoms (or ions).**

Examples of Coordinate Covalent Bond:



METALLIC BOND

A type of bonding which holds the atoms together in metal crystal. A metallic bonding is the electrostatic force of attraction.

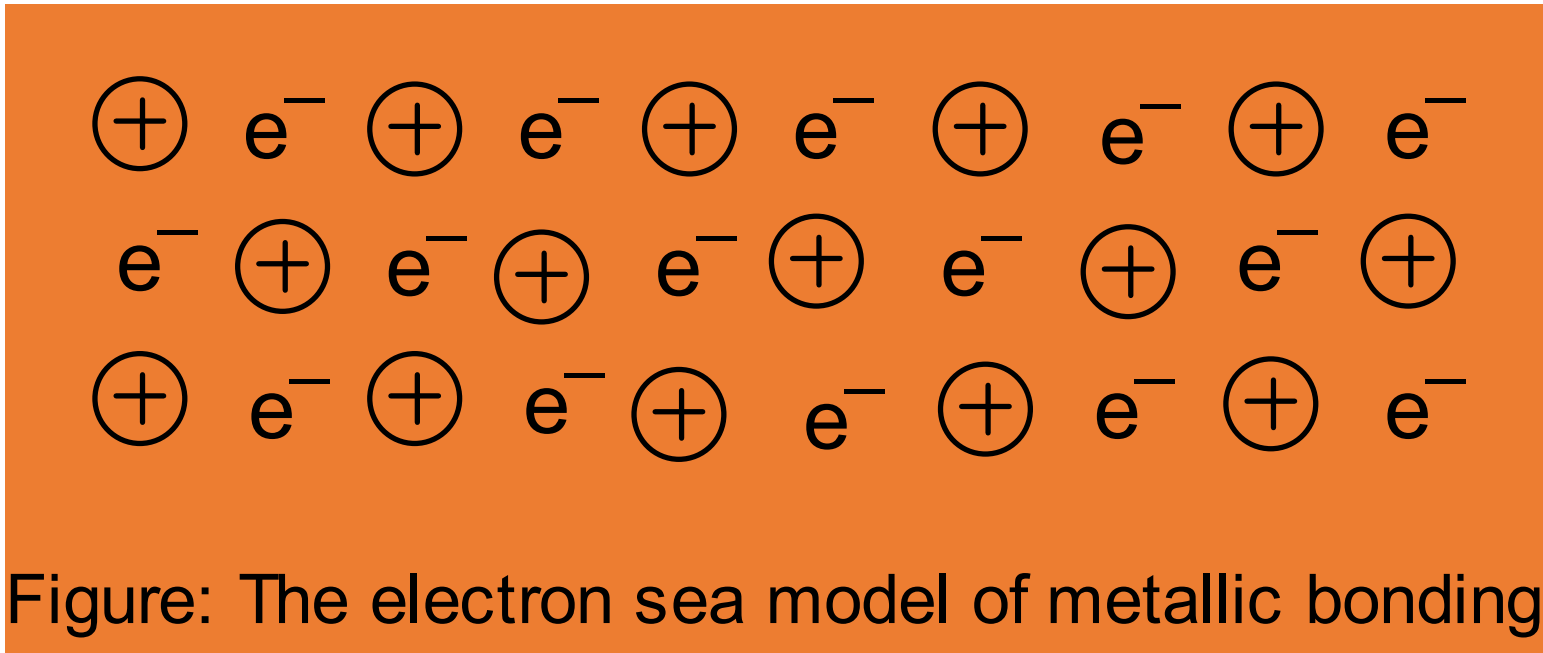
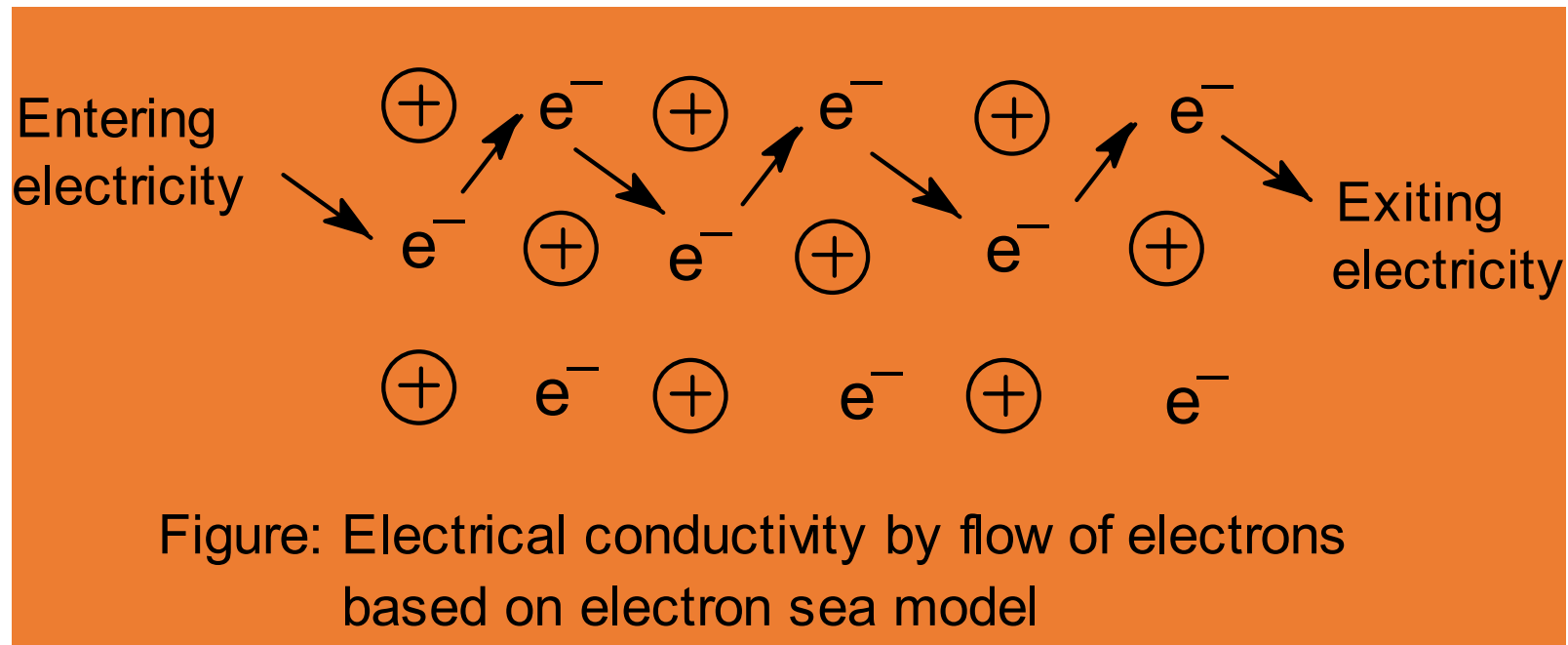


Figure: The electron sea model of metallic bonding

CHARACTERISTICS OF METALLIC BOND

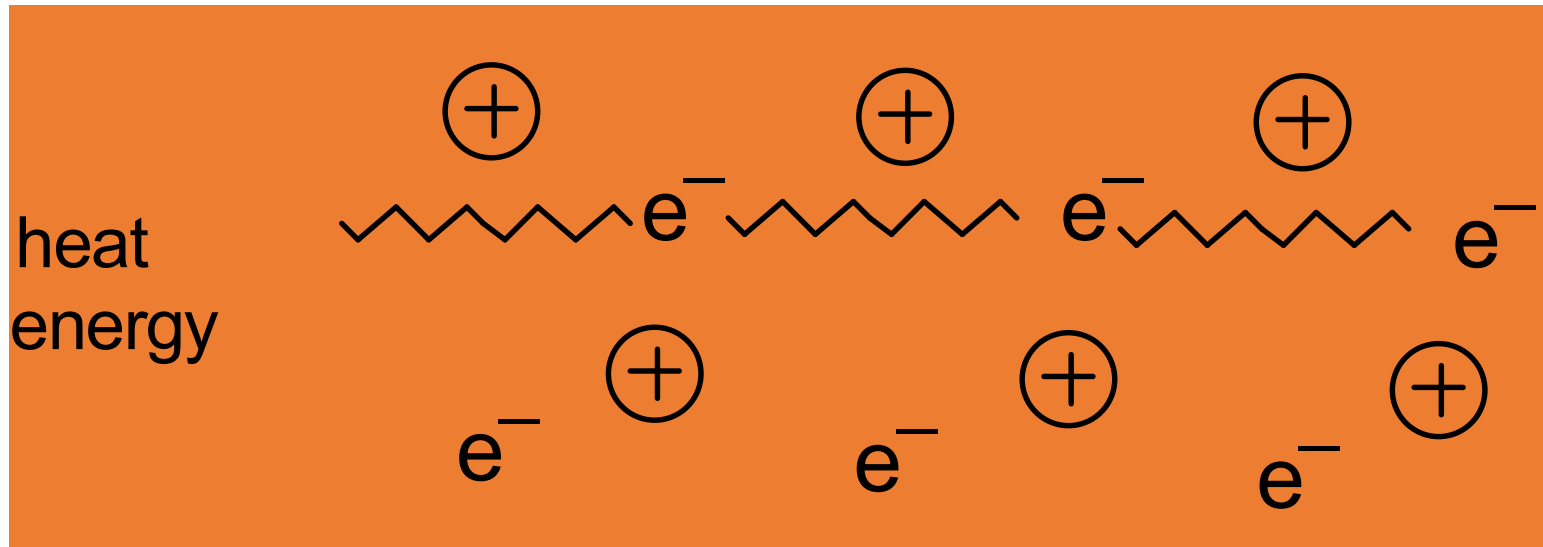
Electric conductivity:



It is attended by electron displacement as a result of voltage difference at both ends. The mobile electrons are responsible for electric conductivity.

CHARACTERISTICS OF METALLIC BOND

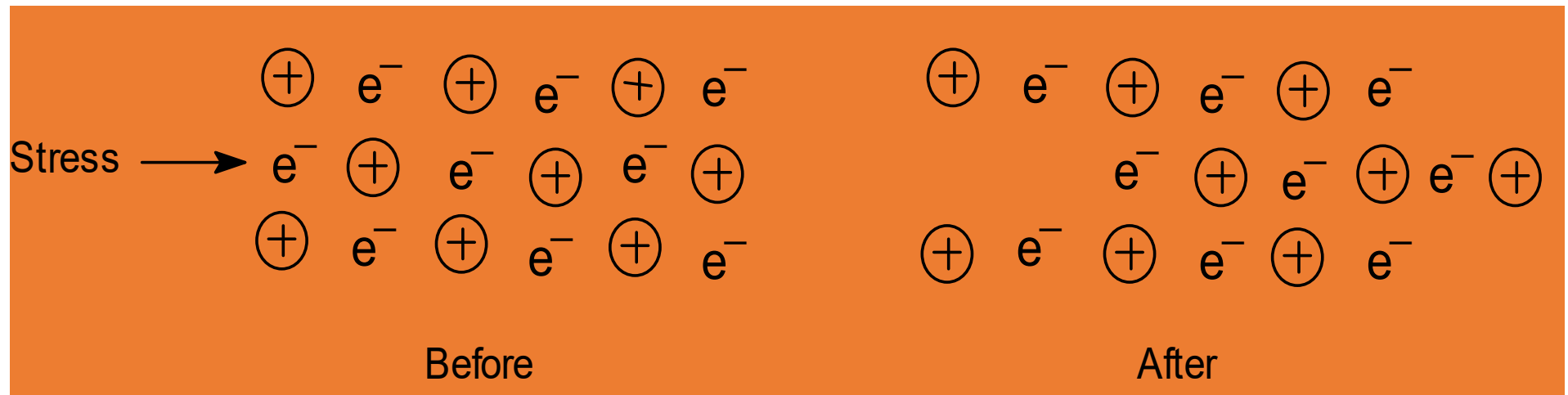
Heat conductivity:



It is done by absorbing heat energy and increasing the vibrational motion of electrons which collide with adjacent electrons and transfer energy.

CHARACTERISTICS OF METALLIC BOND

Ductility and Malleability :



Ductility (*that can be made into thin wire*) and **Malleability** (*can be pressed into different shapes without breaking*): The phenomenon of restoring crystal lattice of metal after applying force, for example by hammering. The sea of electrons adjust position rapidly and the internal structure remains unchanged.

Van der Waals force

- **Three attractive forces exist between neutral molecules:**
- **dipole-dipole forces**
- **London (or dispersion) forces**
- **and hydrogen bonding forces**

The term van der Waals forces is a general term for those intermolecular forces that include dipole-dipole and London forces.

Van der Waals forces are the weak attractive forces in a large number of substances, including Ne, Cl₂, and Br₂

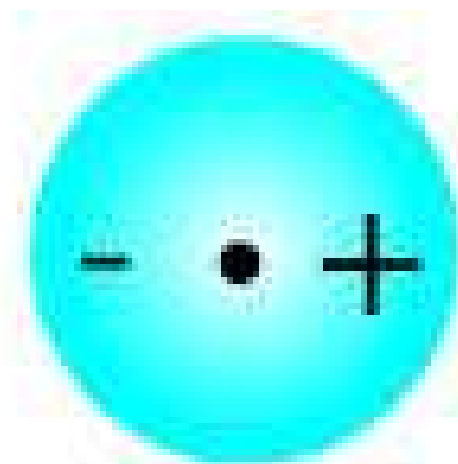
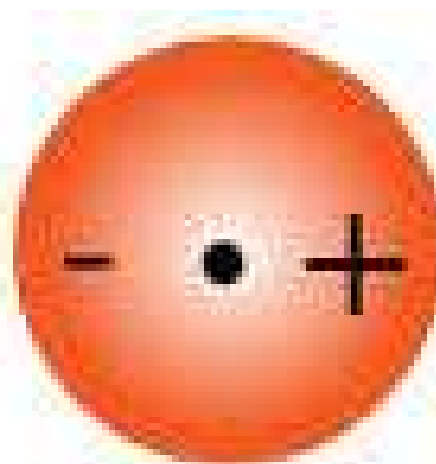
Dipole-Dipole Forces

- **The dipole–dipole force is an attractive intermolecular force resulting from the tendency of polar molecules to align themselves such that the positive end of one molecule is near the negative end of another.**
- **For example, hydrogen chloride, HCl, is a polar molecule because of the difference in electronegativities of the H and Cl atoms.**



London (Dispersion) Forces

- **London forces (also called dispersion forces) are the weak attractive forces between molecules resulting from the small, instantaneous dipoles that occur because of the varying positions of the electrons during their motion about nuclei.**



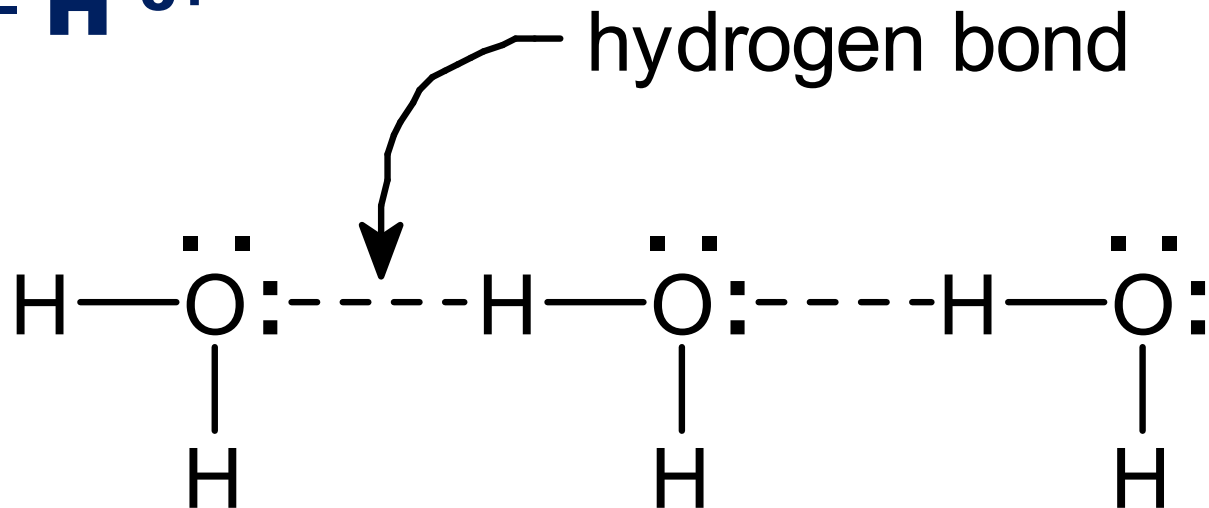
Uneven distribution of
electrons in He

Instantaneous
dipole

Induced dipole
on neighboring He

Hydrogen Bond

- **When hydrogen (H) is covalently bonded to a highly electronegative atoms (X= O, F, Cl), the shared electron pair is pulled so close to X that a strong dipole results.**
- **$X^{\delta-} : - H^{\delta+}$**

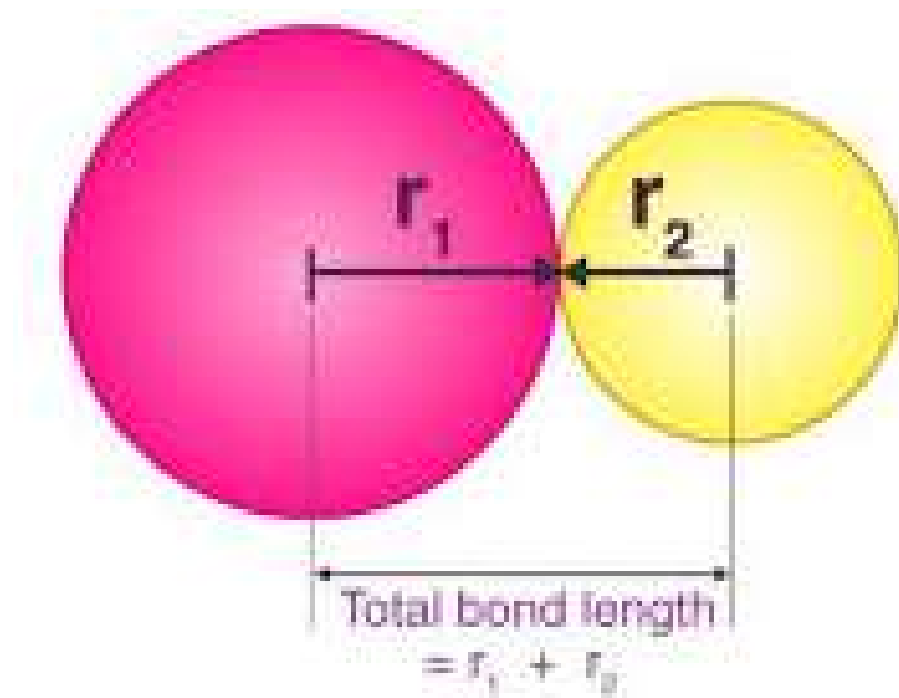


Liquid water

Chemical Bonds

BOND LENGTH

- The distance between the centers of two nuclei of atoms connected by a chemical bond is known as '*bond length*'. It depends on number of bonds. When number of bonds increases, bond length decreases due to more attraction.

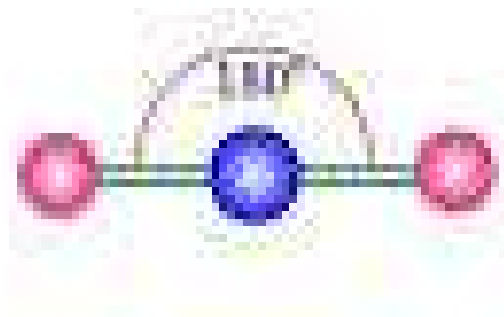


BOND ANGLE

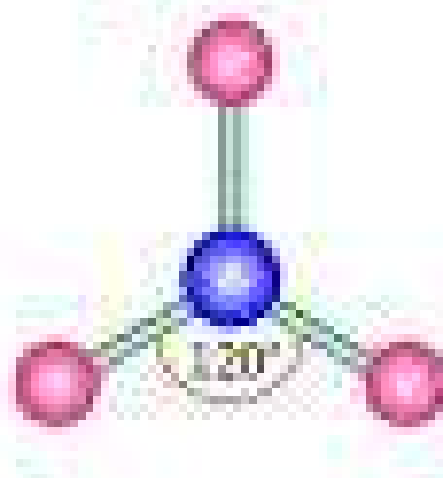
- **The angle between the directions of two bonds in a molecule is called the '*bond angle*'.**
- **Bond angle depends upon**
 - (i) charge distribution**
 - (ii) geometry of the molecules**
 - (iii) symmetry**
 - (iv) hybridization etc.**

BOND ANGLE

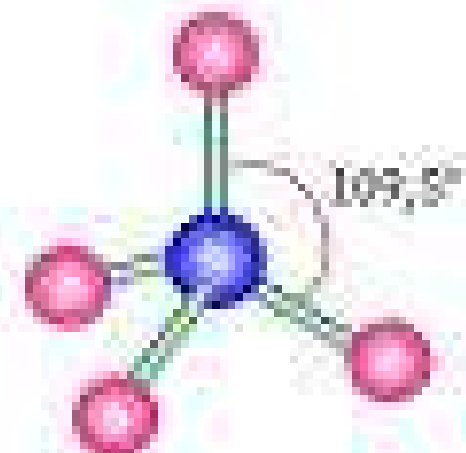
Linear



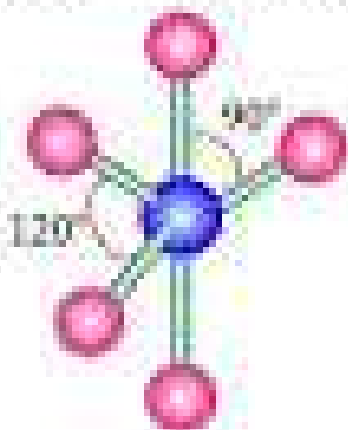
Trigonal Planar



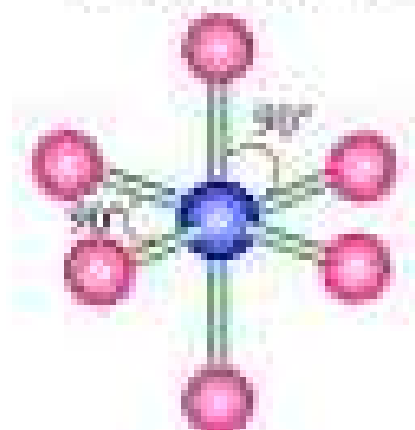
Tetrahedral



Trigonal Bipyramidal

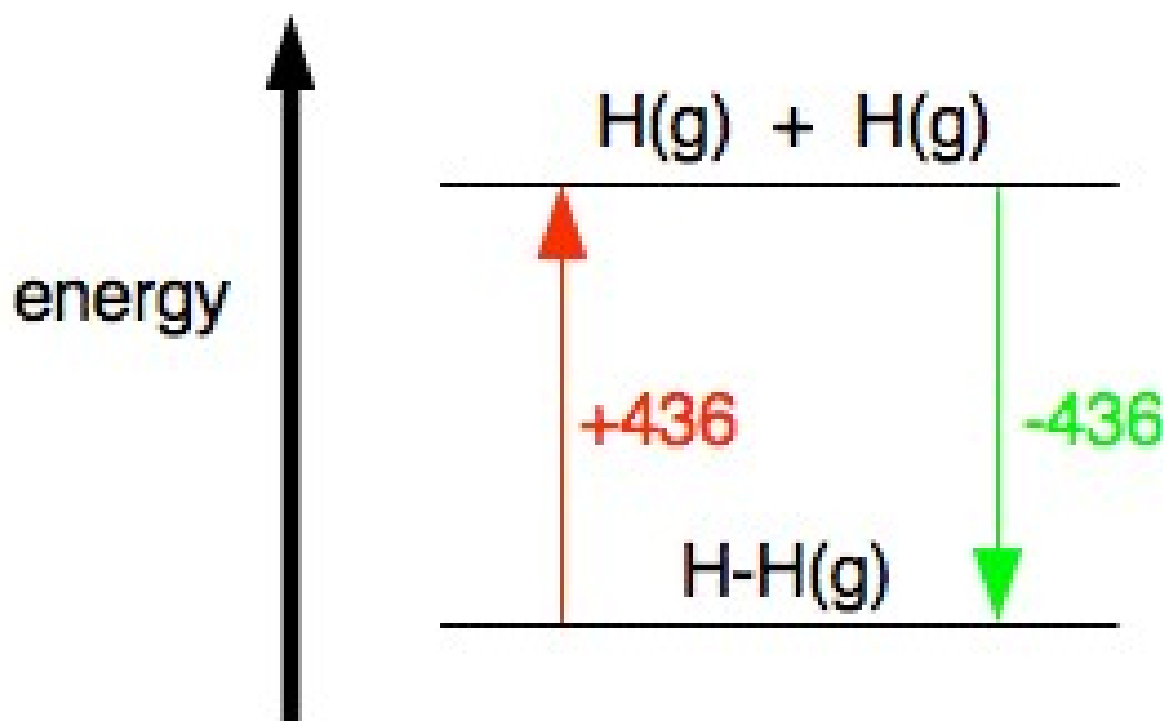


Octahedral



BOND ENERGY

Bond energy is defined as the energy required to split the molecule into atoms. It depends on types of bonding- single, double or triple bonds.



THANK YOU FOR YOUR ATTENTION